

Abderrahim Latreche

alatreche@smu.edu | github.com/rahimcantcode | linkedin.com/in/abderrahimlatreche | +1-469-922-5033

EDUCATION

- **Southern Methodist University - North Texas Scholar** Dallas, TX
Bachelor of Science in Computer Science (AI/ML Track) and Data Science — GPA: 3.9/4.0 *Graduated Dec 2025*
Master of Science in Computer Science (AI/ML Specialization) *Expected Dec 2026*

EXPERIENCE

- **Data Science Intern** Dallas, TX
CBRE *Jun 2025 – Aug 2025*
 - Designed and deployed an ML-driven semantic matching pipeline using retrieval-augmented generation (RAG) and vector embeddings to align natural-language HVAC fault rules with equipment metadata, improving rule coverage across enterprise portfolios
 - Cleaned and transformed **5M+** multivariate HVAC time-series records to engineer features and labeled datasets that powered scalable development of fault-detection machine learning models
 - Codified domain-expert rules into supervised learning targets and model evaluation metrics, benchmarking multiple classifiers against rule-based baselines to quantify predictive lift and assess deployment viability
- **Data Engineer** Plano, TX
Toyota Financial Services *Aug 2025 – Present*
 - Developed a full-stack self-service data portal using React, Python, and REST APIs, enabling **100+** analysts to securely run, schedule, and export SQL queries without engineering support
 - Optimized ETL pipelines and redesigned Snowflake/Postgres schemas, improving query performance by **35** and increasing data reliability across multi-terabyte datasets
- **Research Assistant – Natural Language Processing** Dallas, TX
Southern Methodist University *May 2025 – Aug 2025*
 - Built an **AI advising assistant** using transformer-based NLP models for intent classification and policy responses
 - Created and cleaned text datasets from advising materials to support classification and retrieval tasks
 - Tested LLM pipelines with fine-tuning and embeddings, evaluating precision, recall, and latency
- **Augmented Reality Development Intern** Dallas, TX
Southern Methodist University *May 2024 – Aug 2024*
 - Built AR features for campus navigation using Unity and AR, enabling interactive, location-based guidance
 - Developed scene interactions, UI elements, and environment tracking components to support real-time AR overlays
 - Tested and optimized AR performance across devices, improving stability, responsiveness, and user experience
- **Mathematics Tutor** Dallas, TX
Southern Methodist University *Aug 2023 – Present*
 - Tutored Calculus I-III and Discrete Math; mentored students in problem-solving strategies and study skills

PROJECTS

- **Custom Logistic Regression with Optimization and Regularization** Python
 - Built a multi-class logistic regression model using steepest ascent, SGD, and Newton's method; **85.3%** accuracy
 - Added L1/L2/ElasticNet regularization and PCA, reducing features from 23 to 10 while preserving **95%** variance; matched scikit-learn performance within 0.5s
- **Wide & Deep Neural Network for Booking Cancellation Prediction** Python, TensorFlow
 - Engineered features and trained wide & deep models on a large dataset to predict hotel booking cancellations
 - Evaluated models using ROC-AUC, revenue-weighted F2, and bootstrapping; performance comparable to optimized MLP baseline

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, SAS, C++, Java, C#, JavaScript/TypeScript, HTML/CSS
- **Tools & Frameworks:** scikit-learn, TensorFlow, Keras, MySQL, Git, Linux, Tableau, AWS, Jira, Snowflake
- **Relevant Coursework:** Machine Learning, Advanced Python, NLP, Cloud Computing, AI Applications, Software Testing & QA, Algorithms, Data Structures, Databases, Statistical Computing